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SECTION 1

TUTORIAL ARTICLE REPORT

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1.0 Introduction

1.1 Background of PT. MPM

Through the article, it shown that PT. MPM is a product packaging manufacturing company in Indonesia. The company produces various kinds of packaging products, such as retail packaging, consumables, and shipping supplies. Besides, PT. MPM handles also both mass production and customized job orders based on the customer demand.

In its business operations, PT. MPM receives sales forecasts from customers in annual, quarterly, and monthly formats. These forecasts are used in different purposes such as planning production activities, managing inventory, and supporting decision-making. Other than that, the company also relies on historical sales data, inventory availability, and production manager input to adjust production planning.

1.2 Overview of Performance Monitoring Dashboard

A performance monitoring dashboard is a system that displays business information such as charts, tables, scorecards, and indicators in visual form. It helps users understand company performance more easily without manually checking many separate reports and avoid wasting the time.

For PT. MPM, the dashboard focuses on two main areas:

- 1. Sales Performance Dashboard**

Sales revenue, total sales, forecast sales, actual sales, customer performance, and product category performance.

- 2. Production Performance Dashboard**

Production performance, such as planned production, actual production, machine performance, availability, quality, and Overall Equipment Effectiveness (OEE).

1.3 Purpose of the System

The purpose of the proposed system is to help PT. MPM monitor and evaluate sales and production performance more effectively. By using a Business Intelligence dashboard, the company can access clear and understandable information to support better business decisions.

The dashboard helps the company:

- monitor sales and production performance
- compare forecasted and actual results
- evaluate key performance indicators

2.0 Main Problem

2.1 Current Business Issues

In fact, PT. MPM currently handles large amounts of data from different departments, such as sales, manufacturing, inventory, warehouse, and production. However, the company does not have a proper system to monitor the performance of its core business processes.

The company needs to analyse sales forecasts, actual sales, inventory availability, machine performance, and production output. Without a centralized monitoring system, top management become more difficult to understand the company's real performance quickly.

2.2 Lack of Performance Monitoring System

The main problem faced is the lack of an appropriate application or system to properly assess and monitor business performance. Therefore, top management may not receive accurate, and clear information.

This affects the company's ability to monitor important questions such as:

- Are actual sales meeting the sales forecast?
- Which product categories generate the highest revenue?
- Is production quality meeting the expected standard?
- Are business operations achieving their KPIs?

2.3 Impact on Management Decision-Making

The absence of a proper performance monitoring system can lead to ineffective decision-making. If top management cannot access accurate and understandable information, they may take longer period to evaluate business performance. Besides, making decisions based on incomplete data will also occur. This reduce the company's competitiveness because management cannot quickly identify performance problems or loss the improvement opportunities.

Main Problem Statement:

PT. MPM lacks an appropriate system to measure, monitor, and evaluate its core business processes, especially sales and production performance. It causing top management faces many difficulties in making effective and timely business decisions.

3.0 Main Solution

3.1 Proposed Business Intelligence Dashboard

The proposed solution is to develop a Business Intelligence (BI) dashboard for PT. MPM. A BI dashboard helps transform raw business data into meaningful visual information. The dashboard can also collect, process, and display data from sales and production activities. This allows top management to monitor business performance in an understandable way.

3.2 Sales Performance Dashboard

The Sales Performance Dashboard is designed to help management analyses sales-related data. It allows users to monitor product sales, customer performance, revenue, total sales, forecast sales, and actual sales.

The dashboard includes filters such as:

• date range	• customer ID
• quarter	• product category

This dashboard allows users to view specific sales information based on their needs.

3.3 Production Performance Dashboard

The Production Performance Dashboard is designed to help management evaluate production performance. It focuses on production KPIs such as:

Planned Production	Machine Performance	Quality
Actual Production	Availability	Overall Equipment Effectiveness

This dashboard helps the company compare planned production with actual production and identify machine or production line performance issues.

3.4 Expected Benefits of the System

The proposed dashboard provides several benefits such as:

- a. improves data visibility
- b. supports faster decision-making
- c. reduces manual report checking
- d. improves sales and production analysis
- e. allows management to compare forecasted and actual performance

Main Solution Statement:

The proposed solution is to develop a BI dashboard that visualizes sales and production performance data. It helps top management monitoring KPIs, comparing differences between planned and actual performance, and lastly making better business decisions.

4.0 User Story

4.1 User Story for Top Management.

As a top management user, I want to see sales and production performance dashboards so that I can monitor the company performance and make better business decisions.

4.2 User Story for Data Analyst

As a data analyst, I want to extract, clean, transform, and upload the sales and production data, so that the dashboard can display accurate and useful information.

4.3 User Story for Sales Department

As a sales department user, I want to update sales information such as customer sales, product category, revenue, and sales forecast, so that the company can monitor the sales performance accurately.

4.4 User Story for Manufacturing Department

As a manufacturing department user, I want to update the production information such as planned production, actual production, performance, and quality, so that the production performance can be monitored effectively.

4.5 User Story for Inventory and Warehouse Department

As an inventory and warehouse user, I want to update product stock information, so that the system can provide accurate inventory data for the production and sales analysis.

5.0 Use Case Diagram

5.1 Actors of the System

Actor	Description
Top Management	Views dashboards and make business decisions.
Data Analyst	Cleans, transforms, and uploads data into the dashboard system.
Sales Department	Provides and updates sales-related data.
Manufacturing Department	Provides and updates production-related data.
Inventory and Warehouse Department	Provides and updates product stock information.

5.2 Use Cases of the System

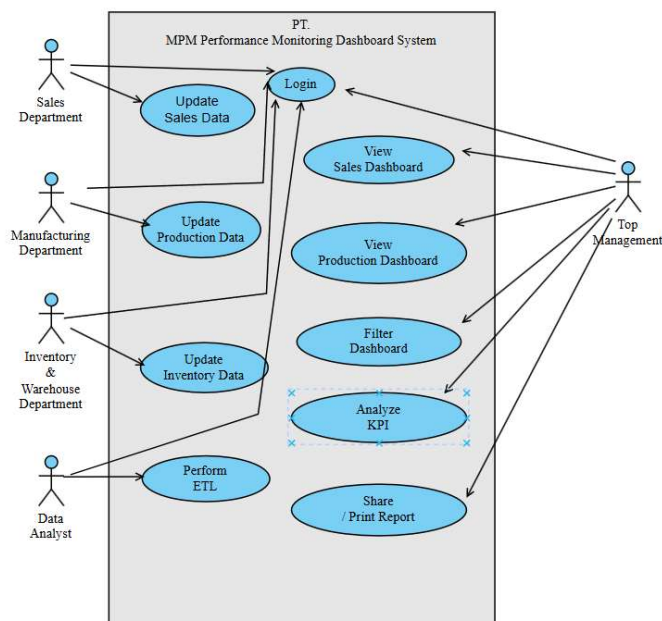


Figure 5.2 PT. MPM Performance Monitoring Dashboard System

5.3 Use Case Diagram Explanation

Actor	Use Case	Explanation
All Users	Login	Users must log in before using the system.

Sales Department	Update Sales Data	Sales staff update sales reports, revenue, customer data, and product category data
Manufacturing Department	Update Production Data	Manufacturing staff update planned production, actual production, machine performance, and quality data.
Inventory & Warehouse Department	Update Inventory Data	Warehouse staff update product stock and inventory information.
Data Analyst	Perform ETL	The data analyst extracts, cleans, transforms, and loads data into the dashboard system.
Top Management	View Sales Dashboard	Top management views sales revenue, total sales, forecast sales, and actual sales.
	View Production Dashboard	Top management views production output, machine performance, quality, and OEE.
	Filter Dashboard	Top management filters data by date, product category, customer, machine, or division.
	Analyze KPI	Top management checks whether sales and

		production performance meet KPI targets.
	Share / Print Report	Top management shares or prints dashboard reports for decision-making.

6.0 Entity Relationship Diagrams

Figure 6.0.1 illustrates the Entity Relationship Diagram (ERD) for the Business Intelligence dashboard of PT. MPM based on Unified Modeling Language (UML) notation.

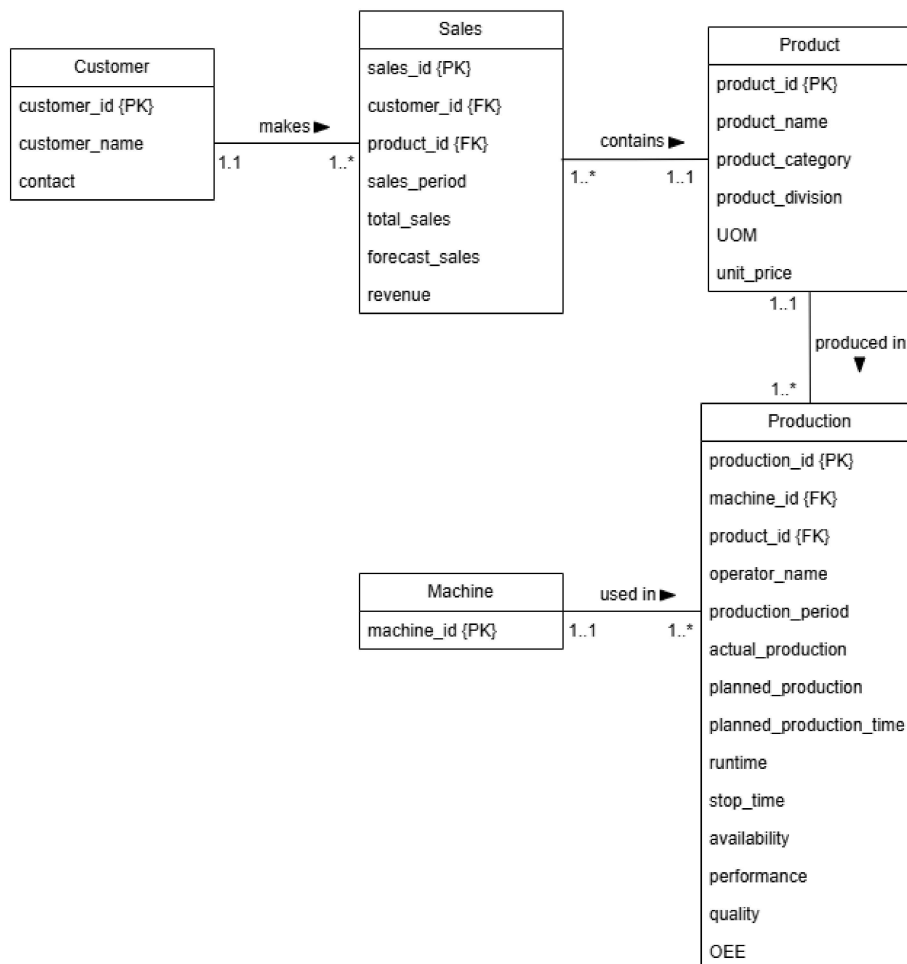


Figure 6.0.1 Entity Relationship Diagram for Business Intelligence Dashboard of PT. MPM

6.1 Identified Entities

The Entity Relationship Diagram (ERD) for the Business Intelligence (BI) dashboard consists of five main entities, representing the core components of the system.

The identified entities are:

- **Customer**

Represents the customer who purchase products. It tracks customer-related sales activities and support sales performance analysis.

- **Product**

Represents the products offered by the company. It stores product-related information and act as a central hub between sales transactions and production processes.

- **Sales**

Represents the sales transactions recorded by the company. It stores sales details such as customer purchases and multiple sales performance metrics.

- **Production**

Represents the production activities carried out by the company. It records production performance data such as the planned and actual production for analysis.

- **Machine**

Represents the machines used in the production processes. It tracks which machine is involved in each production activity.

6.2 Entity Attributes

Each entity in the Entity Relationship Diagram (ERD) consists of a set of attributes that describe its properties. Primary keys (PK) are used to uniquely identify each record, while foreign keys (FK) are used to establish relationships between entities.

Table 6.2.1 explains the attributes in each entity.

Table 6.2.1 Entities' Attribute Explanation Table

Entity	Attribute	Explanation
Customer	customer_id (PK)	Unique identifier for each customer
	customer_name	Name of customer
	contact	Customer contact details
Product	product_id (PK)	Unique identifier for each product
	product_name	Name of product
	product_category	Category of product
	product_division	Product division classification
	UOM	Unit of measurement for product
	unit_price	Selling price per unit
Sales	sales_id (PK)	Unique identifier for each sales record
	customer_id (FK)	References the Customer entity
	product_id (FK)	References the Product entity
	sales_period	Sales time period
	total_sales	Total quantity sold
	forecast_sales	Predicted sales value
	revenue	Total revenue generated
Machine	machine_id (PK)	Unique identifier for each machine
Production	production_id (PK)	Unique identifier for each production record
	machine_id (FK)	References the Machine entity
	product_id (FK)	References the Product entity
	operator_name	Name of the operator involved
	production_period	Time period of production
	actual_production	Actual production quantity
	planned_production	Planned production quantity
	planned_production_time	Planned time for production
	runtime	Actual operating time
	stop_time	Downtime during production
	availability	Machine availability rate
	performance	Machine performance efficiency
	quality	Product without defect

	OEE	Overall Equipment Effectiveness score (availability * performance * quality)
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6.3 Relationships Between Entities

The relationships between the entities define how the data is connected within the system. Each relationship is supported by foreign keys and follows appropriate cardinality.

1. Customer - Sales Relationship (One-to-Many)

- One customer can make many sales transactions.
- Each sales transaction belongs to one customer.

2. Sales – Product Relationship (One-to-Many)

- One product can appear in many sales records.
- Each sales transaction contains one product.

3. Product – Production Relationship (One-to-Many)

- One product can be produced many times.
- Each production record is associated with one product.

4. Machine – Production Relationship (One-to-Many)

- One machine can perform many production activities.
- Each production record uses one machine.

6.4 ERD Explanation

The Entity Relationship Diagram (ERD) is designed to support the Business Intelligence (BI) dashboard in analyzing the sales and production performance.

The Customer entity stores customer information and is connected to the Sales entity to track customer purchases. The Sales entity records sales data such as total sales, forecast sales, and revenue, and links customers with products.

The Product entity stores product details and acts as a central entity because it is connected to both Sales and Production. This allows the system to compare what is sold and what is produced.

The Production entity stores production data such as planned and actual production, as well as performance metrics like runtime, availability, quality, and OEE. These metrics change for each production period, so they are stored in this entity.

The Machine entity represents the machines used in production. Each production record is linked to one machine, while a machine can be used in many production activities.

Overall, the ERD ensures that data is well organized, avoids duplication, and supports accurate analysis for decision-making.

7.0 System Architecture

7.1 Data Sources

The proposed business intelligence (BI) system collects operational data from several departments within PT.MPM. These data sources provide the necessary information for performance monitoring and decision-making.

The sales department provides sales-related information, including product categories, product names, customer information, sales quantity, revenue records and sales forecasts. Forecast data can be provided by year, quarter or month.

The manufacturing department provides production-related data, including machine information, planned production, actual production, operation time, downtime, availability, performance, quality ratio and overall equipment effectiveness indicators.

The inventory and warehouse department provides inventory status, inventory availability and warehouse records to ensure that production activities are consistent with available resources and materials.

7.2 ETL Process

Since the operational data comes from different departments and may contain inconsistent formats, the extraction, transformation and loading process is required to prepare for the analysis of the data.

Extraction

The raw data is collected from sales reports, production reports, maintenance reports and inventory records.

Transformation

The transformation stage consists of several activities to improve data quality and consistency. Data cleaning is performed to remove duplicate, incomplete or irrelevant records. Data standardization is to convert data into a consistent format, such as converting text-based date field into standardize date format. While data integration combining data from different departments into a unified structure suitable for analysis.

Loading

After cleaning and transforming, the data is loaded into the centralized data warehouse repository for storage and analysis.

7.3 Data Warehouse

The Data Warehouse is a centralized repository used to store integrated and cleaned data from multiple operational departments. It provides a single source of truth for organizational reports and business intelligence activities. The data warehouse stores sales data, forecast data, production materials, maintenance data and inventory data. Google sheets are used to organize and maintain the transformed datasets before they are connected to the BI dashboard platform. The warehouse structure supports multi-dimensional analysis and enables efficient reporting for management purposes.

7.4BI Dashboard Layer

The visualization layer is developed using Google Looker Studio. The layer establishes a connection with the data warehouse and converts the raw data into meaningful visual insights. The two dashboards were developed as part of the layer.

The sales performance dashboard allows the management to monitor revenue performance, product performance, customer performance, forecast and actual sales comparison, and sales trends over a period of time.

On the other hand, the production performance dashboard allows management to monitor overall equipment efficiency (OEE), availability, performance and quality indicators, planned and actual production, machine performance and production productivity. Both dashboards provide interactive components such filters, scorecards, charts and heat maps to support data exploration and decision-making.

7.5System Architecture Explanation

The proposed system architecture follows the structure data pipeline approach. First of all, operational data is collected from multiple departments, including sales, manufacturing, inventory and warehouser. Then, the data analyst starts the ETL process to extract, clean, transform and integrate the collected data. After processing, the data is stored in the centralized data warehouse repository. Then, the BI dashboard layer retrieves the processed data and presents it through interactive visualization. These dashboards enable senior managers to monitor business performance, evaluate key performance indicators, identify operational problems and support strategic decision-making.

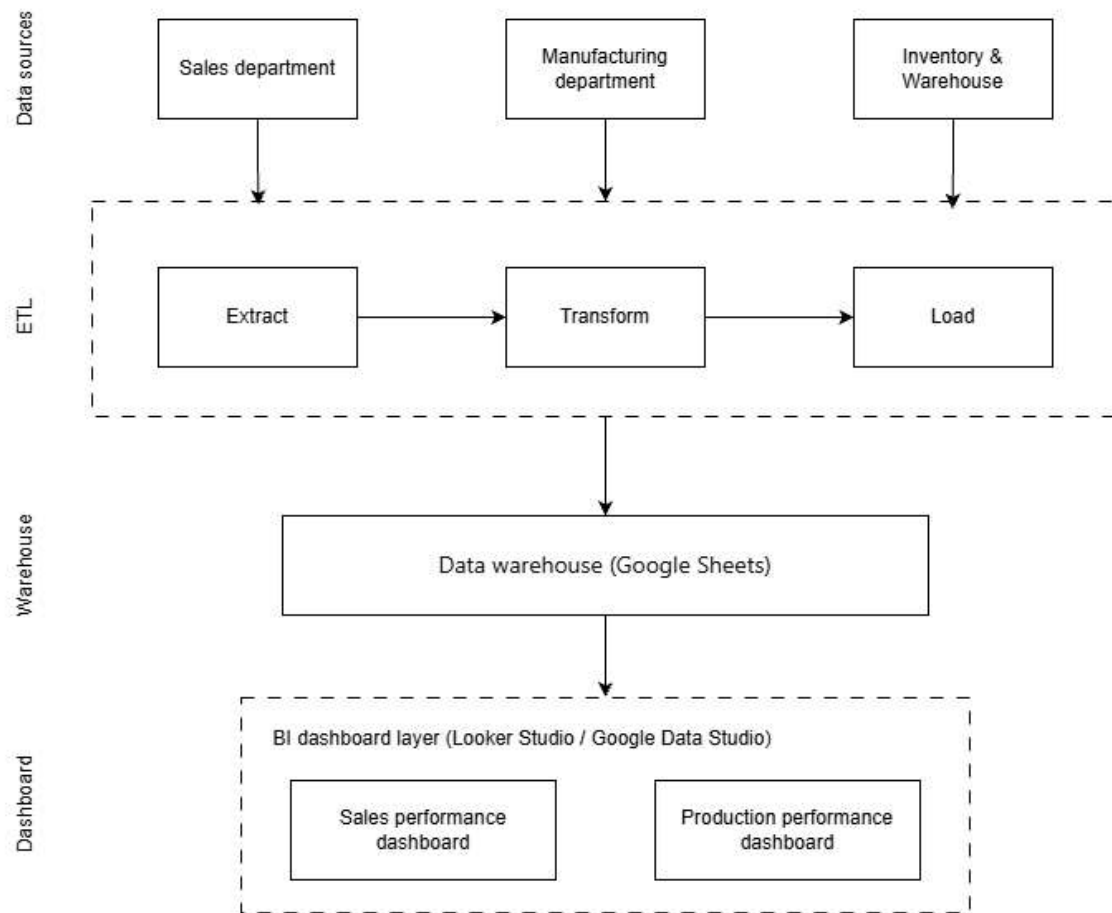


Figure 7.0.1 System architecture

8.0 Multidimensional Model / Star Schema

8.1 Data Dictionary of Galaxy Schema Component Details

In order to support comprehensive business analysis of sales and manufacturing operations, Galaxy schema has been developed. The architecture consists of two central fact tables that share consistent time dimensions, allow management to monitor productivity, OEE and sales performance in a unified data model.

Fact Sales

Field	Type
customer_id	FK
product_id	FK
time_id	FK

quantity_sold	Measure
totalSales	Measure
revenue	Measure
forecastSale	Measure
forecastRevenue	Measure

Fact_Production

Field	Type
machine_ID	FK
time_ID	FK
plannedProduction	Measure
actualProduction	Measure
plannedProductionTime	Measure
runtime	Measure
stopTime	Measure
availability (%)	Measure
performance (%)	Measure
qualityRatio (%)	Measure
OEE (%)	Measure

Dim_Product

Field	Type
product_id	PK
productName	Attribute
productCategory	Attribute

Dim_Customer

Field	Type
customer_id	PK
customerName	Attribute

Dim_Time

Field	Type
time_id	PK
date	Attribute
month	Attribute
quarter	Attribute
productionPeriod	Attribute

Dim_Machine

Field	Type
machine_id	PK
machineName	Attribute
productDivision	Attribute

8.2 Fact Tables and Dimension Tables

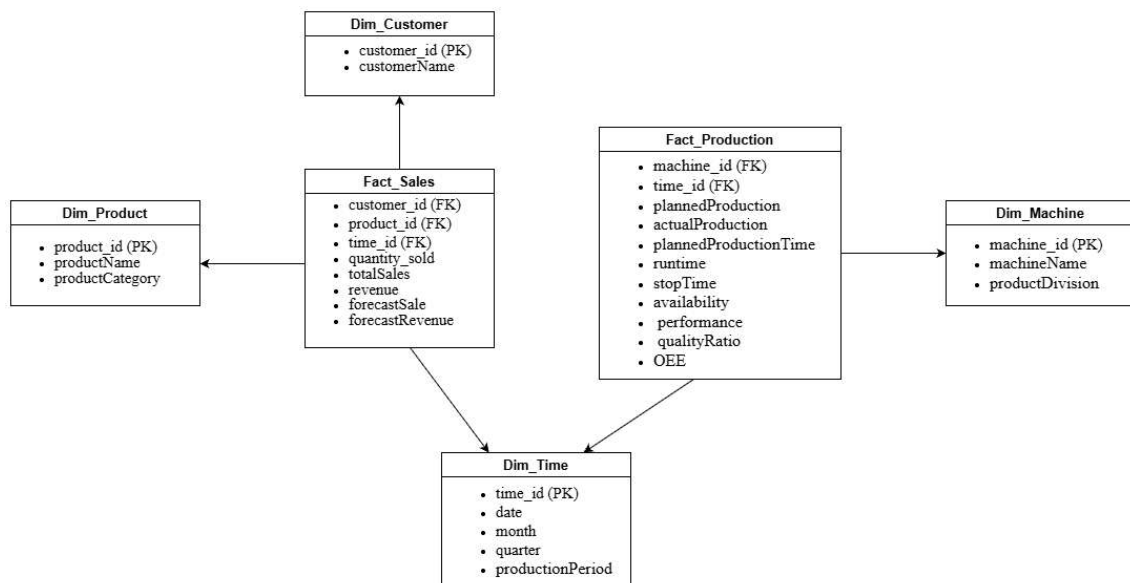


Figure 8.0.1 Sales and Production Performance Galaxy Schema

The fact table stores quantitative business measurements that can be summarized and analyzed, while dimensional table provides descriptive information for filtering, grouping and reporting.

The Fact_Sales table stores sales-related indicator such as revenue and forecast value. It is related to product, customer and time dimensions.

The Fact_Production table stores manufacturing performance indicators such as production count, execution time, stop and OEE measurement. It is associated with the machine and the time dimensions.

Through this integrated architecture, both fact tables share the matching Dim_Time dimension to achieve comprehensive cross-functional business analysis.

8.3 Galaxy Schema Explanation

The proposed multi-dimensional model is designed with a galaxy schema because it support multiple business processes in a single integrated data warehouse while maintaining efficient query performance. Similar to the star schema, the dimensional table is directly connected to the fact table without additional normalization layer. This design reduces the complexity of the query, minimizes table connections and improves the response time of the dashboard.

For the sales performance dashboard, management can analyze revenue, sales and forecast achievements by filtering data by product categories, customers, and time periods. For the production performance dashboard, the management can evaluate production efficiency by comparing the plan and the actual production output and monitor machine performance through the overall equipment efficiency (OEE) indicators. OEE combines usability, performance and quality measurement to conduct a comprehensive evaluation of manufacturing efficiency.

Galaxy Schema enables Google Data Studio to efficiently retrieve and visualize data from multiple business areas, supporting real-time performance monitoring and strategic decision-making.

9.0 References

Jayadinata, D. H., Pratama, A. T., & Sofianti, T. D. (2024). Development of Performance Monitoring Dashboard for Product Packaging Manufacturer by using Waterfall

Methodology. *Jurnal Ilmiah Teknik Industri*, 151–161.

<https://doi.org/10.23917/jiti.v23i1.2551>